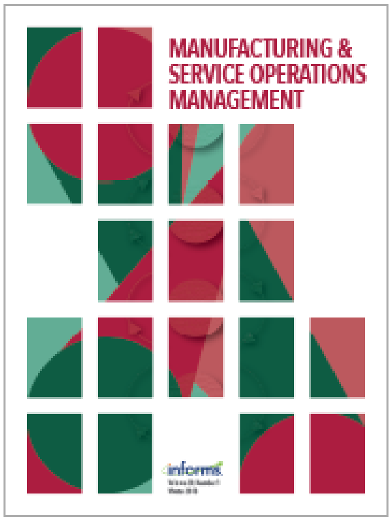

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Balancing Supply with Demand on Ride-Hailing Platforms in Markets with Price Regulations: An Operational Approach

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Abstract. ***Problem definition*:** Ride-hailing platforms face a supply-demand imbalance. During peak periods, the demand from passengers far exceeds the supply of drivers, whereas during off-peak periods, there is an abundance of supply but weak demand. The well-studied surge pricing can be challenging to implement in markets where prices are subject to regulation and are inflexible to adjust. We study an alternative operational approach that shifts the supply of drivers from off-peak to peak periods to address the supply-demand imbalance. ***Methodology/results*:** We propose two novel incentive schemes: the *qualification* scheme and the *prioritization* scheme. Specifically, the platform sets a work target for the peak period. Under the qualification scheme, the platform assigns off-peak service requests only to drivers who meet the peak period target. Under the prioritization scheme, the platform prioritizes off-peak requests for drivers who meet the peak period target. We analyze the effectiveness of these schemes, considering the openness of the platform's supply system. For platforms with a closed system that only allows full-time drivers to provide service, the qualification scheme improves the total matching volume to a greater extent but hurts full-time drivers more than the prioritization scheme. For platforms with an open system that also allows the abundant part-time drivers to serve off-peak requests, the prioritization scheme outperforms the qualification scheme in improving total matching volume. Furthermore, the implementation of an incentive scheme in an open system may benefit both the platform and full-time drivers. ***Managerial implications*:** Our study suggests an alternative to surge pricing for on-demand platforms to address the supply-demand imbalance when prices are inflexible. It provides insights for platforms with varying levels of supply system openness in choosing appropriate incentive schemes. The welfare results offer guidance for platforms and policymakers regarding both matching volume and worker welfare.

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Keywords: ride-hailing platforms • supply-demand imbalance • operational approach • incentive schemes

1. Introduction

On-demand ride-hailing platforms, such as Didi, connect passengers with on-call drivers. A distinguishing feature of these two-sided markets is that the drivers are typically independent contractors rather than formal employees of the platform, giving them flexibility in choosing their work amount and schedule. However, as drivers' choices may not always align with the platform's interests, significant discrepancies between service supply and demand usually occur and bring great challenges to platforms. For example, Didi, the Chinese ride-hailing giant, constantly faces labor shortages during weekday mornings when requests for rides peak, and the resulting unfulfilled ride requests account for one-fifth of the total demand (Shen 2019). In cities such as Beijing, the likelihood of securing a ride is lower during peak periods compared with normal periods (Wei 2019). One way to ease the supply-demand imbalance is surge pricing, which involves raising the price to attract more drivers during peak hours to meet the increased demand for rides. Nevertheless, in markets such as China, where ride-hailing prices are regulated and less flexible,[1] implementing surge pricing can be challenging. Since 2017, Didi has faced multiple lawsuits for increasing prices by 20% during peak periods, with accusations of abusing market power (Wang 2019).

Moreover, surge pricing itself has been criticized by passengers who feel exploited and view the practice as unfair and sometimes unethical. It may also hurt drivers. Using field data from one ride-hailing platform in China, Xu et al. (2024) show that customers may misdirect complaints about surge pricing toward individual drivers, increasing the likelihood of drivers receiving complaints by a factor of 1.13.

In response to regulations on price control and surge pricing limitations, some Chinese platforms are turning to nonpricing strategies to address the supply-demand imbalances. For example, Licheng Zhuanche, a platform focusing on premium service, allocates off-peak ride requests only to drivers whose service levels during peak periods meet the platform's requirements. This approach aims to incentivize drivers to provide more services during peak periods (Sohu 2021). Didi, on the other hand, implements the "reputation score" system in 2019, where each driver's score is positively correlated with their amount of service at peak hours. During the off-peak periods, those with high scores will be assigned ride requests with high priority.[2] Both Licheng Zhuanche and Didi's approaches do not rely on varying the price to adjust demand but focus on changing service incentives to shift supply.

Motivated by the emerging practices, this paper studies two novel incentive schemes based on supply shifting and examines their effectiveness in solving the supply-demand imbalance and influence on platform profitability and users' welfare. In addition to the peak periods when the increase in demand causes a shortage of service supply, there are also off-peak periods when service supply exceeds demand. For example, drivers in Shanghai struggle to receive ride orders during off-peak times (OFWeek 2023). It is common for drivers to be idle during these periods (Feng et al. 2025). The two schemes, namely the *qualification* scheme, resembling Licheng Zhuanche's approach, and the *prioritization* scheme, similar to Didi's approach, aim to shift the supply of service from off-peak periods to peak periods without altering the transaction price. This is achieved by strategically assigning off-peak service requests to drivers based on their peak period work amount. Specifically, the platform sets a target level of service for the peak period. In the *qualification* scheme, during off-peak periods, the platform assigns service requests only to drivers who have provided service up to the work target during peak periods. This means that drivers who have provided less service during peak periods are not qualified to work during off-peak periods. In the *prioritization* scheme, the platform first assigns service requests in off-peak periods to drivers who have achieved the target during peak period and then assigns the remaining requests to other drivers. This prioritizes drivers who have met the target during peak periods for work during off-peak periods.

To examine the effectiveness of the incentive schemes, it is essential to understand the two main reasons for supply shortage during peak periods. The first reason is related to the composition of the service supply. For example, most Didi drivers are part-time, and half of them work fewer than two hours a day (Liao 2019, Liu and Lu 2021). Only 16.76% of Didi drivers work more than six hours a day (Liu and Lu 2021). As a result, many part-time drivers may not be able to work during peak periods because of their full-time jobs, causing a peak-period supply shortage.[3] The second reason stems from the worse traffic conditions, which lead even full-time drivers to be less motivated to work during peak periods despite the excessive demand. Multiple studies have consistently shown that traffic congestion (Stokols et al. 1978, Gulian et al. 1989, Hennessy and Wiesenthal 1997, Evans et al. 2002) causes increased driver stress, along with the emotional discomfort of working at rush hours (Morris and Hirsch 2016). In addition, drivers often receive unfair low ratings from passengers because of factors beyond their control, such as traffic-related delays, further diminishing their motivation to provide service during the peak periods (Kapoor and Tucker 2017, Ashkrof et al. 2020). As a result, although full-time drivers can provide service during peak periods, they may reduce the service amount. In this study, we capture both causes of the peak period supply shortages and examine how qualification and prioritization schemes can shift service supply without changing the transaction price.

When studying the two incentive schemes, we also consider the openness of a platform's supply system. Some platforms adopt an *open* system, allowing both full-time and part-time workers to provide services. Other platforms adopt a *closed* system by excluding drivers who cannot provide service full-time. For example, Caocao Chuxing, a new entrant to China's ride-hailing market, enforces strict work amount requirements that effectively deter part-time drivers from joining the platform (Goh and Shirouzu 2018, Tang 2018). Since January 2019, Didi has been transitioning from an open system to a closed system by implementing double licensing, requiring both drivers and their cars to be licensed for service provision. As registering a privately owned car for commercial use entails significant costs in China (Liao 2019, Nonninger 2019), the double licensing policy essentially excludes part-time drivers from supplying services. We show that the openness of a platform's supply system is crucial in determining the two incentive schemes' effectiveness.

To focus attention on the operational approach and reflect the price's inflexibility, we assume that both periods have the same price per unit of service time, which is exogenously given. We begin with a closed system in which part-time drivers are excluded from the service supply, and we study a benchmark scenario in which full-time drivers freely allocate their total work time across peak and off-peak periods.

Working during the peak period incurs an additional congestion cost, which reflects the psychological discomfort, high stress levels, and fatigue experienced by drivers. In deciding how to allocate their work hours, full-time drivers weigh the increased likelihood of receiving ride requests against the added congestion cost of working during peak periods. We find that the optimal allocation of work hours by full-time drivers may not align with the platform's best interests, as it does not lead to maximizing the total matching volume.

We then study the two incentive schemes and show that both schemes can improve the platform's total matching volume compared with the benchmark scenario. Furthermore, in choosing the peak period target for both schemes, the platform chooses the smallest target that can induce all full-time drivers to participate, rather than choosing a larger target that only involves some of the drivers in the scheme. Given that both schemes engage all full-time drivers while excluding part-time drivers from the service supply, one may believe that the two schemes are equivalent for a platform with a closed system. Interestingly, we find that the qualification scheme leads to a higher total matching volume than the prioritization scheme. This is because, if not reaching the peak period work target, full-time drivers cannot provide service during the off-peak period under the qualification scheme but can still receive ride requests under a prioritization scheme (although with a lower priority). This difference enables the platform to set a higher work target under a qualification scheme, shift more supply from the off-peak period to the peak period, and improve the total matching volume to a greater extent. Despite the increased matching volume, full-time drivers are worse off because of the longer work time during the peak period.

We next turn the focus to an open system where a significant number of part-time workers can provide service during the off-peak period. Compared with a closed system, full-time drivers under an open system allocate more of their service time to the peak period to avoid intense competition with part-time drivers in the off-peak period, resulting in a less severe peak period supply shortage. Similar to a closed system, for both schemes, the platform chooses an optimal peak period work target that involves all full-time drivers in the scheme. However, in contrast to a closed system, we find that a prioritization scheme outperforms a qualification scheme in improving the total matching volume. The prioritization scheme offers two key advantages over the qualification scheme. First, it effectively utilizes the service supply from part-time drivers, resulting in improved off-peak matching volume. Second, the prioritization scheme allows the platform to increase the peak period work target by leveraging the abundant off-peak supply from part-time drivers, thus improving the peak period matching volume. We also show that compared with the benchmark scenario without any scheme, the introduction of an incentive scheme in an open system may increase the platform's total matching volume and benefit full-time drivers simultaneously.

In extensions, we compare our proposed schemes with surge pricing in various performance measures and discuss the situations where each approach can be effectively applied. We then relax the model assumption by allowing a finite number of part-time drivers in an open system and explore the robustness and boundary conditions of our main results.

The remainder of the paper proceeds as follows. After reviewing the literature in Section 2, we present the model in Section 3. Sections 4 and 5 analyze the benchmark scenario and the two schemes in a closed system and an open system, respectively. Section 6 extends the model to compare with surge pricing and to assess the results' robustness. Finally, Section 7 concludes the paper. All proofs are presented in the Online Appendix and supplementary analysis.

2. Literature Review

The recent development of on-demand service platforms has drawn significant attention in academia. A growing body of literature has investigated various aspects of on-demand service platforms, especially operational issues on ride-hailing platforms. Examples of such issues include dynamic pricing and capacity management (Cachon et al. 2017, Taylor 2018, Bai et al. 2019, Yang et al. 2020, Hu et al. 2022), matching demand with supply (Bimpikis et al. 2019, Besbes et al. 2021, Hu and Zhou 2022), and labor welfare, government regulation, and gender-based issues (Yu et al. 2020, Tang et al. 2021, Besbes et al. 2024). We refer interested readers to Narasimhan et al. (2018), Wang and Yang (2019), and Benjaafar and Hu (2020) for reviews of studies on these topics.

Our paper contributes to the literature on using nonpricing instruments to facilitate service provision. Yang and Tang (2018) introduce a new fare-reward scheme for railway transit that aims to shift commuters' peak departures to off-peak periods. By incorporating the heterogeneity in commuters' scheduling flexibility, Tang et al. (2019) further propose a hybrid fare scheme that combines the fare-reward scheme with a nonrewarding scheme to shift peak traffic to off-peak periods. Yang et al. (2020) build a reward scheme integrated with a constrained surge pricing and show that the novel approach can simultaneously benefit users, drivers, and the platform. Afèche et al. (2023) consider the interplay of passengers' admission control and drivers' repositioning decisions over a spatial network and find that the platform may reject requests that originated from low-demand regions even when the supply is abundant so that drivers will self-relocate to high-demand locations. Different from the existing studies that largely focus on adjusting the demand for service, we fix the service demand but explore the shift of service supply through operational approaches.

Our paper also relates to the management of service providers' participation. Hu et al. (2022) study a platform's worker classifications. They show that compared with uniformly classifying both long-term workers and ad hoc workers, discriminatorily classifying long-term workers as employees and ad hoc workers as contractors can mitigate undercutting and overjoining issues and lead to a Pareto improvement outcome. Castro et al. (2022) study a platform's supply prioritization over private agents, who are fully controlled by the platform and receive a fixed wage, and flexible agents, who freely determine to provide service and pay a commission to the platform. They show that, depending on whether the market is over- or undersupplied, the platform should prioritize either the private or the flexible agents. Besbes et al. (2024) propose a novel prioritized-first-come-first-serve (PFCFS) policy, allowing drivers to reserve slots according to their temporal preferences to raise drivers' effective wage without hurting supply. Our study differs from existing research that relies on bonus incentives, information provision, worker classification, or supply prioritization to manage service providers. Instead, we consider the context when service providers are heterogeneous in their working schedules, and we focus on shifting self-scheduling capacity from off-peak periods to peak periods through operational approaches to address the supply-demand imbalance. Our paper contributes to the literature by characterizing the optimal incentive scheme in the presence of heterogeneous service providers.

3. Model Setup

Consider an on-demand ride-hailing platform that receives ride requests from passengers and assigns them to available drivers. The transaction price of a trip typically depends on various factors such as the distance traveled, duration of the trip, and the number of baggage items. To keep the model focused, we assume the transaction price to be proportional to the service time, and we denote the price per unit of service time as p. When a transaction occurs, the platform pays a proportion δ of the total transaction price to the driver while retaining the remaining $1-\delta$ as its own revenue. To explore the use of an operational approach for balancing supply and demand, we treat the price per unit service time as exogenous. The assumption of exogenous pricing aligns with existing literature that explores nonpricing factors on two-sided platforms (Feng et al. 2021, Afèche et al. 2023) and is applicable for modeling markets with strong price regulation. In Section 6.1, we compare the performance of our operational approach with surge pricing.

Each passenger participates in the market at a length-less time point and sends her service request to the platform. The demand for service fluctuates significantly throughout the day. During rush hours, which we refer to as the peak period, the service demand far exceeds the supply capacity from drivers, leading to significant supply shortage. Conversely, there are also hours, such as early morning and late night, when supply exceeds demand, which we refer to as the off-peak period.

3.1. Drivers

We consider two types of drivers. Some drivers can freely choose to work during both peak and off-peak periods and are referred to as full-time drivers in our model. We use n to denote the size of the full-time drivers. Each full-time driver allocates his total work time, which is normalized to one, between the peak and the off-peak periods. We use $\alpha \in [0,1]$ and $1-\alpha$ to denote the proportions of work time that a full-time driver allocates to the peak and the off-peak periods.[4] Evidenced by multiple studies (Hennessy and Wiesenthal 1999, Morris and Hirsch 2016, Huang et al. 2018) and as discussed in Section 1, drivers incur additional costs, such as psychological discomfort, high stress levels, and fatigue, when working during the congested peak periods. These additional costs could cause full-time drivers to be less willing to work during the peak period, especially when price is not adjusted. To reflect this, we normalize the cost of working during the off-peak period to zero and introduce an additional congestion cost of $(1/2)k \cdot \alpha^2$ for working during the peak period. The quadratic cost function is used to capture the increasing marginal cost associated with the working time during the peak period. The parameter k measures the magnitude of the additional costs. It is important to emphasize that the additional costs are borne on drivers but not on the platform. As a result, drivers' decisions on how to allocate their working time may not align with the platform's optimal interests.

In addition to full-time drivers, some platforms also involve drivers who are only available to work in the off-peak period because of their primary jobs or other commitments during peak hours. We refer to the drivers who are constrained to work in the off-peak period as part-time drivers. To capture the supply surplus during the off-peak period in the simplest way and to maintain analysis tractability and to focus attention on shifting full-time drivers' supply from the off-peak period to the peak period, we assume that the number of part-time drivers is infinite. The assumption is purely for technical tractability. In Section 6.2, we consider the case of finite part-time drivers and demonstrate the robustness of our main results.

In practice, there may also be drivers who are constrained to work during the peak period. As long as the additional service supply does not address the peak period's supply shortage, the presence of these drivers would not affect a full-time driver's time allocation between peak and off-peak periods and therefore does not qualitatively affect our main results on implementing the incentive schemes.[5] To maintain simplicity, we do not explicitly model this group of drivers.

3.2. The Supply-Demand Imbalance

For simplicity, we only consider the total demand of each period, which is understood as the integration of the number of passengers in each specific time point over the total duration of that period. We denote $D_p(p)$ as the peak period demand given price p and assume $D_p(p) > n$, reflecting the peak period supply shortage even when full-time drivers devote all their time to peak periods.[6] We use $D(\epsilon) = D_0 - p + \epsilon$ to denote the total service demand in the off-peak period, where ϵ is a mean-zero random factor to reflect the off-peak demand randomness and follows a distribution F, which is differentiable and has the support $[-s, s]$. We define $L \equiv D_0 - p$ and assume $s \in (0, \min\{n - L, L\})$. Here, the constraint $s < n - L$ implies $D(\epsilon) = L + \epsilon < L + n - L = n$, indicating that service demand falls short of supply during the off-peak period when all full-time drivers devote all their time to the off-peak period. The constraint $s < L$ ensures that the realized off-peak demand $D(\epsilon)$ is always positive.

Lastly, if $k \le (np\delta)/(n - L + s)$ (i.e., the congestion cost is very small), the service target might be lower than the amount a full-time driver would voluntarily work without any incentive scheme, making an incentive scheme irrelevant. Therefore, we constrain our analysis to the region $k > (np\delta)/(n - L + s)$. This region corresponds to a scenario where the driver congestion cost is not too small, which discourages drivers from working a long time in the peak under fixed pricing and presents an opportunity for our incentive schemes to effectively incentivize drivers to shift their availability from off-peak periods to peak hours and address the supply-demand imbalance.

3.3. Platform, Incentive Schemes, and System Openness

We start with a benchmark model in which the platform randomly assigns ride-hailing requests to all available drivers. We then analyze two incentive schemes, namely, the *qualification* scheme and the *prioritization* scheme, with the goal of shifting service supply from the off-peak period to the peak period to solve the supply-demand imbalance. Specifically, the platform publicly sets a peak period work target α_0. Under the *qualification* scheme, the platform assigns off-peak period requests only to drivers who have fulfilled the peak period target amount. Under the *prioritization* scheme, the platform first assigns off-peak requests to drivers who have fulfilled the peak period target and then assigns the residual requests, if any, to other drivers.

The openness to service supply varies across platforms. Some platforms allow both full-time and part-time drivers to provide service, which we call an *open* system. Other platforms only allow full-time drivers to provide service, which we refer to as a *closed* system. In practice, the service supply openness may depend on various factors, including concerns regarding transaction security, the platform's control over independent drivers, and the regulatory environment. For the purpose of this research, we take the system openness as exogenous and study how the effectiveness of the two incentive schemes in improving matching volume depends on system openness. Table 1 summarizes our key notation.

4. Closed System

A platform with a closed system allows only full-time drivers to provide services. For succinctness, we use drivers to refer to full-time drivers in this section. We start by examining a benchmark case in which no incentive scheme is implemented. Then, we show the impact of qualification and prioritization schemes on drivers' work time allocation, welfare, and the total matching volume.

4.1. Benchmark

We first derive the driver's equilibrium work time allocation when no incentive scheme is implemented. Recall that α is the proportion of time that a full-time driver allocates to the peak period. As demand exceeds supply in the peak period, a driver can always receive ride requests and earn $p \cdot \delta$ income for each unit of time. In the meantime, the driver also incurs the congestion cost $(1/2)k \cdot \alpha^2$. Therefore, the driver's net utility from working during peak periods is $p\delta \cdot \alpha - (1/2)k \cdot \alpha^2$.

We next consider the off-peak period and use $\hat{\alpha}$ to denote all drivers' average amounts of work time allocated to the peak period in equilibrium. The total service supply during the off-peak period is $n(1 - \hat{\alpha})$. Each driver can serve a passenger with certainty if $D(\epsilon) \ge n(1 - \hat{\alpha})$ but otherwise serve a passenger with probability $\frac{D(\epsilon)}{n(1-\hat{\alpha})}$. Therefore, a driver's net utility from working during the off-peak period is $\min\left\{1, \frac{D(\epsilon)}{n(1-\hat{\alpha})}\right\} \cdot p\delta \cdot (1 - \alpha)$. Summing up a driver's utility from both

Table 1. Summary of Key Notation

	Notation	Explanation
The exogenous parameters	p	Transaction price per unit of service time, $p \in (0, D_0)$
	n	The size of the group of full-time drivers
	D_0	The intercept of the off-peak demand function, $D_0 \in (0, n+p-s)$
	L	Expected demand during off-peak periods, $L \equiv D_0 - p$
	ϵ	Uncertain component of off-peak demand, $\epsilon \in [-s, s]$, $s \in (0, \min\{n-L, L\})$
	δ	The proportion of transaction price retained by the driver, $\delta \in (0,1)$
	k	Congestion cost coefficient, $k \in \left(\frac{np\delta}{n-L+s}, +\infty\right)$
The platform's decision	α_0	Peak period work target, $\alpha_0 \in (0,1)$
A full-time driver's decision	α	The proportion of time allocated to the peak period, $\alpha \in [0,1]$
The endogenous parameter	θ	The proportion of qualified/prioritized (full-time) drivers
Superscripts	B	Benchmark
	Q	Qualification scheme
	P	Prioritization scheme
Subscripts	q/uq	Qualified/unqualified (full-time) drivers
	p/up	Prioritized/unprioritized (full-time) drivers
	C	Closed system
	O	Open system

periods, we obtain the driver's utility function for a given realized ϵ as shown below, where the superscript B and subscript C refer to the benchmark and the closed system, respectively:

$$U_C^B(\alpha,\epsilon) = p\delta\cdot\alpha - \frac{1}{2}k\cdot\alpha^2 + \min\left\{1, \frac{D(\epsilon)}{n(1-\hat{\alpha})}\right\}\cdot p\delta\cdot(1-\alpha). \tag{1}$$

A driver chooses α to maximize the expected utility $U_C^B(\alpha) = \int_{-s}^{s} U_C^B(\alpha,\epsilon)dF(\epsilon)$. Lemma 1 characterizes a full-time driver's optimal choice of α, the resulting utility, and the platform's expected total matching volume.

Lemma 1 (Closed System: Benchmark). *In the closed system and without an incentive scheme, the equilibrium is characterized as follows:*

- *Each full-time driver's optimal choice of α, denoted by α_C^B, is given as*

$$\alpha_C^B = \begin{cases} \alpha^b, & \text{if } k \le k_1, \\ \dfrac{k + p\delta - \sqrt{(k-p\delta)^2 + 4\frac{kL}{n}p\delta}}{2k}, & \text{if } k > k_1, \end{cases}$$

where α^b solves the implicit function $\alpha^b = \frac{p\delta}{k}\cdot\frac{\int_{-s}^{n(1-\alpha^b)-L}F(\epsilon)d\epsilon}{n(1-\alpha^b)}$ and $k_1 = \frac{p\delta s}{(L+s)(1-\frac{L+s}{n})}$.

- *Each full-time driver's utility, denoted by U_C^B, is given as*

$$U_C^B = \begin{cases} \left(1 - \dfrac{\int_{-s}^{n(1-\alpha^b)-L}F(\epsilon)d\epsilon}{n}\right)p\delta - \dfrac{1}{2}k{\alpha^b}^2, & \text{if } k \le k_1, \\ \dfrac{-k^2 + 2\frac{kL}{n}p\delta + 2\delta kp + \delta^2p^2 + (k-p\delta)\sqrt{(k-p\delta)^2 + 4\frac{kL}{n}p\delta}}{4k}, & \text{if } k > k_1. \end{cases}$$

- *The platform's expected total matching volume across both periods, denoted by M_C^B, is*

$$M_C^B = \begin{cases} n - \displaystyle\int_{-s}^{n(1-\alpha^b)-L}F(\epsilon)d\epsilon, & \text{if } k \le k_1, \\ L + n\dfrac{k + p\delta - \sqrt{(k-p\delta)^2 + 4\frac{kL}{n}p\delta}}{2k}, & \text{if } k > k_1. \end{cases}$$

Furthermore, we have $M_C^B < n$.

When allocating their work time between the two periods, a full-time driver weighs two opposing effects. On one hand, working during the off-peak period exposes them to increased competition and the risk of receiving fewer ride requests. On the other hand, working during the peak period means incurring the additional cost of congestion despite the presence of ample demand. Lemma 1 shows that the total expected matching volume across both periods is always less than the total capacity (i.e., $M_C^B < n$). This is because full-time drivers, because of the high congestion costs in the peak period, tend to allocate more service to the off-peak period than what is demanded during that time. As a result, there is an excess of available capacity during the off-peak period, leading to idleness and underutilization of resources.

We next analyze the two proposed incentive schemes to study their effectiveness in increasing the matching volume.

4.2. Qualification Scheme

When adopting the *qualification* scheme, the platform assigns off-peak ride requests exclusively to drivers who have met the peak period work target. Therefore, for a given peak period work target α_0, each driver decides whether to participate in the scheme by allocating

$\alpha \geq \alpha_0$ amount of time to the peak period. The platform, anticipating how each driver will respond to the target service amount, chooses the work target that maximizes the total matching volume. We first derive each driver's choice of work time allocation given the platform's choice of α_0 and then determine the platform's optimal choice of α_0.

To facilitate analysis, we define $\theta \in (0, 1]$ as the proportion of drivers qualified to provide service during the off-peak period. Given our model of a continuum of drivers, each driver's unilateral decision to participate or not in the scheme will not affect θ. We then specify a driver's utility of participating or not in the qualification scheme.

- If a driver does not participate in the qualification scheme, he can only provide service during the peak period. The unqualified driver, by working α_{uq} amount of time during the peak period, derives a utility of $U_{uq}(\alpha_{uq}) = p\delta \cdot \alpha_{uq} - (1/2)k \cdot \alpha_{uq}^2$. Using the first-order condition, we can solve for an unqualified driver's choice of α_{uq}, denoted by α_{uq}^Q, as $\alpha_{uq}^Q = p\delta/k$. The corresponding driver utility is $U_{uq}^Q = p^2\delta^2/(2k)$.
- If a driver participates in the qualification scheme, he needs to work at least α_0 during the peak period to be qualified to provide service in the off-peak period. We use α_q to represent the driver's work time during the peak period. Given the equilibrium average supply decision $\hat{\alpha}_q$ of all qualified drivers, the total service supply in the off-peak period becomes $n\theta(1 - \hat{\alpha}_q)$. Thus, a qualified driver's utility for a realized ϵ is

$$U_q(\alpha_q, \epsilon) = \min\left\{1, \frac{D(\epsilon)}{n\theta(1 - \hat{\alpha}_q)}\right\} \cdot p\delta \cdot (1 - \alpha_q) + p\delta \cdot \alpha_q - \frac{1}{2}k \cdot \alpha_q^2, \tag{2}$$

and a qualified driver chooses α_q to maximize the expected utility

$$\max_{\alpha_q} \; U_q(\alpha) = \max_{\alpha_q} \left(\int_{-s}^{s} U_q(\alpha_q, \epsilon) dF(\epsilon) \right), \tag{3}$$

$$\text{s.t.} \;\; \alpha_0 \leq \alpha_q \leq 1. \tag{4}$$

The equilibrium condition requires $\alpha_q = \hat{\alpha}_q$. Our next result characterizes each qualified driver's equilibrium choice of α_q, denoted as α_q^Q.

Lemma 2. *Given a closed system, under a qualification scheme with $\alpha_0 > \frac{p\delta}{k}$, each qualified driver chooses $\alpha_q^Q = \alpha_0$.*

To understand Lemma 2, note that a qualification scheme only becomes relevant if the work target exceeds what drivers would voluntarily choose when not participating in the scheme (i.e., $p\delta/k$). Drivers who participate in the scheme only work up to the platform's target in the peak period to qualify for off-peak work, as additional peak period work beyond the target would only incur congestion costs without providing extra benefits.

Anticipating each type of driver's choice of $(\alpha_q^Q, \alpha_{uq}^Q)$, the platform chooses the target α_0, which induces θ proportion of full-time drivers to voluntarily participate, to maximize the expected total matching quantity $M^Q(\theta(\alpha_0))$. The matching volume during the off-peak period equals the smaller of riding demand and service supply, that is, $\min\{D(\epsilon), n\theta(1 - \alpha_0)\}$. The peak period matching volume is determined by the service supply (i.e., $n\theta\alpha_0 + n(1 - \theta)\alpha_{uq}^Q$) because of the supply shortage. Thus,

$$M^Q(\theta(\alpha_0)) = \int_{-s}^{s} \min\{D(\epsilon), n\theta(1 - \alpha_0)\} dF(\epsilon) + n\theta\alpha_0 + n(1 - \theta)\alpha_{uq}^Q. \tag{5}$$

The platform's problem can be formulated as

$$\max_{\alpha_0} \; M^Q(\theta(\alpha_0)), \tag{6}$$

$$\text{s.t.} \; U_{uq}^Q \leq U_q(\alpha_0) \text{ with the constraint binding if } \theta < 1. \tag{7}$$

Constraint (7) is the incentive compatibility condition. It suggests that each driver must be indifferent between being qualified and unqualified if there is a mix of both types of drivers. If all drivers voluntarily choose to be qualified, each driver derives weakly higher utility than not being qualified. Lemma 3 characterizes the equilibrium under the qualification scheme.

Lemma 3 (Closed System: Qualification Scheme). *In the closed system with a qualification scheme, the equilibrium is characterized as follows:*

- *The platform's optimal choice of work target, denoted by α_0^Q, is given as*

$$\alpha_0^Q = \begin{cases} 1 - \dfrac{L - s}{n}, & \text{if } k \leq k_2, \\ \alpha_0^q, & \text{if } k_2 < k \leq k_3, \\ \dfrac{p\delta}{k} + \sqrt{2\dfrac{L}{n}\dfrac{p\delta}{k}}, & \text{if } k > k_3, \end{cases}$$

where α_0^q is larger than $\frac{p\delta}{k}$ and solves the implicit function $\left(1 - \frac{1}{n}\int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon\right)p\delta - \frac{1}{2}k\alpha_0^{q\,2} = \frac{p^2\delta^2}{2k}$, $k_2 = \frac{p\delta\left(\sqrt{\frac{L-s}{n}\left(2-\frac{L-s}{n}\right)}+1\right)}{\left(1-\frac{L-s}{n}\right)^2}$, and $k_3 = \frac{p\delta\left(\sqrt{\frac{L}{n}\left(2-\frac{L+2s}{n}\right)}-\frac{s}{n}+1\right)}{\left(1-\frac{L+s}{n}\right)^2}$.

- *Given α_0^Q, all drivers participate in the scheme (i.e., $\theta(\alpha_0^Q) = 1$). Each full-time driver's utility, denoted by U_C^Q, is given as*

$$U_C^Q = \begin{cases} p\delta - \dfrac{1}{2} \cdot k\left(1 - \dfrac{L - s}{n}\right)^2, & \text{if } k \leq k_2, \\ \dfrac{p^2\delta^2}{2k}, & \text{if } k > k_2. \end{cases}$$

• *The platform's expected total matching volume across both periods, denoted by* M_C^Q, *is*

$$M_C^Q = \begin{cases} n, & \text{if } k \le k_2, \\ n - \displaystyle\int_{-s}^{n(1-\alpha_0^q)-L} F(\epsilon)d\epsilon, & \text{if } k_2 < k \le k_3, \\ L + n\dfrac{p\delta}{k} + \sqrt{2nL\dfrac{p\delta}{k}}, & \text{if } k > k_3. \end{cases}$$

Lemma 3 shows that the platform maximizes the total matching volume by inducing all drivers to participate in the qualification scheme. That is, compared with setting a high target and only engaging some drivers in the scheme, the platform can improve the matching volume by lowering the target and engaging as many drivers as possible. Note that the capacity shifted to the peak period is fully utilized because of high demand, whereas the same capacity may be idle during the off-peak period because of the limited and stochastic demand. By having all drivers work a relatively longer time during the peak period, it not only increases the peak period matching volume but also alleviates competition during the off-peak period, thereby increasing overall capacity utilization.

The platform's choice of peak period work target critically hinges on the congestion coefficient k. Given that $k > (np\delta)/(n - L + s)$, if the congestion cost is small (i.e., $k \le k_2$), the platform is able to raise the work target ($\alpha_0^Q = 1 - (L - s)/n$), which shifts a large amount of service supply from the off-peak period to the peak period. The significant supply shift also fully eliminates the capacity idleness in the off-peak period, resulting in full-capacity utilization ($M_C^Q = n$). However, as the congestion cost rises beyond a threshold (i.e., $k > k_2$), the platform has to lower the peak period work target to ensure that all drivers participate in the qualification scheme. As a result, the off-peak supply may still exceed the off-peak demand, which implies that the total matching volume, although improved compared with the benchmark case, could still be lower than total capacity.

4.3. Prioritization Scheme

We next turn our attention to the *prioritization* scheme, under which the platform also announces a peak work target α_0. Unlike a qualification scheme, drivers who work less than α_0 during the peak period are still able to provide service during the off-peak period but are not given priority in being assigned ride requests. Specifically, the platform assigns off-peak period requests first to drivers who have worked up to α_0 during the peak period and only assigns any remaining requests to the remaining drivers.

We follow the same steps as in Section 4.2 by first deriving each driver's work time allocation given a certain work target α_0 and then solving for the platform's optimal choice of α_0. We still use $\theta \in (0,1]$ to denote the proportion of drivers who participate in the scheme and are prioritized in being assigned off-peak period requests. The remaining $1-\theta$ proportion of drivers are unprioritized in being assigned off-peak period requests.

• If a driver does not participate in the prioritization scheme, he works $\alpha_{up} \in [0, \alpha_0)$ amount of time during the peak period and derives utility $p\delta \cdot \alpha_{up} - (1/2)k \cdot \alpha_{up}^2$ from the peak period. In the off-peak period, only if $D(\epsilon) - n\theta(1-\hat{\alpha}_p) > 0$ (or equivalently, $\epsilon > n\theta(1-\hat{\alpha}_p) - L$), unprioritized drivers can receive service requests, as off-peak demand cannot be fully fulfilled by prioritized drivers. An unprioritized driver's chance of receiving a request is $\min\{1, \frac{D(\epsilon)-n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}\}$, where $\hat{\alpha}_p$ and $\hat{\alpha}_{up}$ are the average peak period supply of all prioritized and unprioritized drivers, respectively. As a result, the unprioritized driver's utility from the off-peak period is $\min\{1, \frac{D(\epsilon)-n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}\} \cdot p\delta \cdot (1-\alpha_{up})$. Combining the utility from both periods, an unprioritized driver's work time allocation problem is

$$\max_{\alpha_{up}<\alpha_0} U_{up}(\alpha_{up}) = \max_{\alpha_{up}} \Bigg(p\delta\alpha_{up} - \frac{1}{2}k\alpha_{up}^2 + \int_{n\theta(1-\hat{\alpha}_p)-L}^{s} \min\left\{1, \frac{D(\epsilon) - n\theta(1-\hat{\alpha}_p)}{n(1-\theta)(1-\hat{\alpha}_{up})}\right\} p\delta(1-\alpha_{up})dF(\epsilon)\Bigg). \tag{8}$$

• If a driver participates in the prioritization scheme, he works $\alpha_p \in [\alpha_0, 1]$ in the peak period. We can similarly formulate his work time allocation problem as

$$\max_{\alpha_p \ge \alpha_0} U_p(\alpha_p) = \max_{\alpha_p} \Bigg(p\delta\alpha_p - \frac{1}{2}k\alpha_p^2 + \int_{-s}^{s} \min\left\{1, \frac{D(\epsilon)}{n\theta(1-\hat{\alpha}_p)}\right\} dF(\epsilon) p\delta(1-\alpha_p)\Bigg). \tag{9}$$

Anticipating the equilibrium work time allocation of a prioritized driver (α_p^*) and that of an unprioritized driver (α_{up}^*), the platform chooses the target α_0 to maximize the total matching volume $M^P(\theta(\alpha_0))$. Note that the total service supply capacities during peak and off-peak periods are $n\theta\alpha_p^* + n(1-\theta)\alpha_{up}^*$ and $n\theta(1-\alpha_p^*) + n(1-\theta)(1-\alpha_{up}^*)$, respectively. Thus, the total matching volume is

$$M^P(\theta(\alpha_0)) = \int_{-s}^{s} \min\{D(\epsilon), n\theta(1-\alpha_p^*) + n(1-\theta)(1-\alpha_{up}^*)\} dF(\epsilon) + n\theta\alpha_p^* + n(1-\theta)\alpha_{up}^*. \tag{10}$$

The platform's problem can be formulated as

$$\max_{\alpha_0} \; M^P(\theta(\alpha_0)), \tag{11}$$

$$\text{s.t. } U_{up}(\alpha_{up}^*) \le U_p(\alpha_p^*) \text{ with the constraint binding if } \theta < 1. \tag{12}$$

Constraint (12) is the incentive compatibility condition that ensures that the proportions θ and $1-\theta$ of drivers self-select to become prioritized and unprioritized drivers. Lemma 4 characterizes the equilibrium.

Lemma 4 (Closed System: Prioritization Scheme). *In the closed system with a prioritization scheme, the equilibrium is characterized as follows:*

- *The platform's optimal choice of work target, denoted by α_0^P, is given as*

$$\alpha_0^P = \begin{cases} \alpha_0^p, & \text{if } k \le k_3, \\ \dfrac{p\delta}{k} + \sqrt{2\dfrac{L}{n}\dfrac{p\delta}{k}}, & \text{if } k > k_3, \end{cases}$$

where α_0^p solves the equation $\left(1 - \frac{1}{n}\int_{-s}^{n(1-\alpha_0^p)-L} F(\epsilon)d\epsilon\right)p\delta - \frac{1}{2}k\alpha_0^{p2} = \frac{p\delta\left(2k\int_{n\left(1-\alpha_0^p\right)-L}^{s} dF(\epsilon) + p\delta\left(1-\int_{n\left(1-\alpha_0^p\right)-L}^{s} dF(\epsilon)\right)^2\right)}{2k}$.

- *Given α_0^P, all drivers participate (i.e., $\theta(\alpha_0^P) = 1$). Each full-time driver's utility is*

$$U_C^P = \begin{cases} \left(1 - \dfrac{1}{n}\displaystyle\int_{-s}^{n(1-\alpha_0^p)-L} F(\epsilon)d\epsilon\right)p\delta - \dfrac{1}{2}k\alpha_0^{p2}, & \text{if } k \le k_3, \\ \dfrac{p^2\delta^2}{2k}, & \text{if } k > k_3. \end{cases}$$

- *The platform's expected total matching volume across both periods, denoted by M_C^P, is*

$$M_C^P = \begin{cases} n - \displaystyle\int_{-s}^{n(1-\alpha_0^p)-L} F(\epsilon)d\epsilon, & \text{if } k \le k_3, \\ L + n\dfrac{p\delta}{k} + \sqrt{2nL\dfrac{p\delta}{k}}, & \text{if } k > k_3. \end{cases}$$

Furthermore, we have $M_C^P < n$.

Similar to the result of a qualification scheme, with a prioritization scheme, the platform also chooses a target α_0^P that induces all drivers to participate. But unlike a qualification scheme, the platform is unable to achieve a full capacity utilization (i.e., $M_C^P < n$), even when the congestion cost is small. To appreciate the intuition, it is important to note that with a prioritization scheme, drivers still have the opportunity to work during the off-peak period despite a lower priority for receiving service requests if they choose not to participate in the scheme. This distinction from a qualification scheme limits the platform's ability to set a high peak work target when engaging all drivers. When the congestion cost is small, a qualification scheme allows the platform to shift enough supply to the peak period and fully eliminate the capacity idleness in the off-peak period, whereas a prioritization scheme cannot achieve this because of the lower work target set by the platform.

Interestingly, when the congestion cost is large (i.e., $k > k_3$), both the prioritization and qualification schemes achieve the same outcomes for the platform's work target, each driver's utility, and the total matching volume (see Lemma 3 and Lemma 4). This convergence occurs because in the presence of high congestion costs, the platform faces limitations in shifting sufficient supply to the peak period to eliminate off-peak capacity idleness, regardless of the scheme employed. Consequently, drivers who do not participate in either scheme are unable to receive service requests during the off-peak period, leading to a driver facing the same trade-off between participating or not in both of the schemes. As a result, the platform's optimal choice of the peak work target and the corresponding driver welfare and transaction volumes are the same across both schemes.

4.4. Comparison of the Benchmark and Two Schemes in a Closed System

We now compare the benchmark scenario with the two incentive schemes to examine their effects on the total matching volume and drivers' welfare.

Proposition 1. *In the closed system, given $k > \frac{np\delta}{n-L+s}$ and that the price per unit of service time is fixed, we have the following results:*

- *Implementing a prioritization scheme improves the platform's total matching volume compared with the benchmark. However, it lowers each driver's welfare.*
- *Implementing a qualification scheme further increases the total matching volume compared with the prioritization scheme. However, it further lowers each driver's welfare.*
- *The platform sets a (weakly) higher peak period work target when employing a qualification scheme as opposed to a prioritization scheme.*

The reason that a driver becomes worse off under a prioritization scheme, despite a higher total matching volume, is that the peak work target set by the platform is higher than that a driver would voluntarily choose (in the absence of any scheme). As the service supply can always be utilized during the peak period but can be wasted during the off-peak period if demand is insufficient, the prioritization scheme increases the total matching volume by shifting supply from the off-peak period to the peak period. However, each individual driver is worse off because of the additional congestion cost caused by the prolonged work time during the peak period.

Recall that under either scheme, the platform uses a peak period work target that induces all drivers to participate in the scheme. One may expect that the two schemes are equally effective in shifting off-peak supply to the peak period. However, as readily indicated by Lemma 3 and Lemma 4, the qualification scheme proves to be more effective in improving the total match volume. If a driver does not meet the peak period work target, he can still work during the off-peak period under the prioritization scheme but is excluded from the service supply under the qualification scheme. As a result, the platform is able to set a higher target under the qualification scheme to shift more service supply from the off-peak period to the peak period and improve the matching volume to a greater extent. However, drivers are further hurt as a result of the longer work time during the peak period.

5. Open System

A platform with an open system allows both full-time and part-time drivers to provide service. In this section, we analyze the benchmark case and the two incentive schemes in an open system and compare their performances.

5.1. Benchmark

When no scheme is implemented and with the presence of the large supply capacity from part-time drivers, a full-time driver's chance of serving a passenger during the off-peak period is negligible. With the simplifying assumption of infinite part-time drivers, a full-time driver derives zero utility from working in the off-peak period. Therefore, a full-time driver's utility when allocating α time to the peak period is

$$U_O^B(\alpha) = p\delta \cdot \alpha - \frac{1}{2}k \cdot \alpha^2. \tag{13}$$

Lemma 5 summarizes the benchmark equilibrium.

Lemma 5 (Open System: Benchmark). *In the open system without an incentive scheme, the equilibrium is characterized as follows:*

- *Each full-time driver's choice of α, denoted by α_O^B, is $\alpha_O^B = \frac{p\delta}{k}$.*
- *Each full-time driver's utility is given as $U_O^B = \frac{p^2\delta^2}{2k}$.*
- *The platform's expected total matching volume across both periods is $M_O^B = L + n\frac{p\delta}{k}$.*

Comparing the matching volume and driver welfare in an open system with those in a closed system, we have the following result:

Proposition 2. *In the absence of an incentive scheme, a platform achieves a higher total matching volume in an open system than in a closed system. However, full-time drivers are worse off.*

An open system improves the number of matching volumes in two ways. Directly, it expands the service supply during the off-peak period by involving part-time drivers. Indirectly, it shifts the full-time drivers' work time more toward the peak period so that the peak period matching volume also increases. However, the full-time drivers are worse off, as they have to work more during the peak period and bear additional congestion costs.

5.2. Qualification Scheme

When the platform adopts a qualification scheme, part-time drivers will not be assigned any service requests, as they do not serve passengers during the peak period. Therefore, the qualification scheme in an open system is effectively the same as that in a closed system. The equilibrium has the same characterization as that in Lemma 3.

5.3. Prioritization Scheme

Following a similar procedure as that for a closed system, we obtain the equilibrium outcomes of using a prioritization scheme in an open system, as shown in Lemma 6.

Lemma 6 (Open System: Prioritization Scheme). *In the open system with a prioritization scheme, the equilibrium is characterized as follows:*

- *The platform's choice of peak period work target, denoted by α_{0O}^P, is*

$$\alpha_{0O}^P = \begin{cases} \sqrt{\frac{p\delta}{k}\left(2 - \frac{p\delta}{k}\right)}, & \text{if } k \le k_2, \\ \alpha_0^q, & \text{if } k_2 < k \le k_3, \\ \frac{p\delta}{k} + \sqrt{2\frac{L}{n}\frac{p\delta}{k}}, & \text{if } k > k_3. \end{cases}$$

- *All full-time drivers participate in the scheme. Each driver's utility is $U_O^P = \frac{p^2\delta^2}{2k}$.*
- *The platform's expected total matching volume, denoted by M_O^P, is*

$$M_O^P = \begin{cases} L + n\sqrt{\frac{p\delta}{k}\left(2 - \frac{p\delta}{k}\right)}, & \text{if } k \le k_2, \\ L + n\alpha_0^q, & \text{if } k_2 < k \le k_3, \\ L + n\frac{p\delta}{k} + \sqrt{2nL\frac{p\delta}{k}}, & \text{if } k > k_3. \end{cases}$$

When a prioritization scheme is implemented, full-time drivers who do not participate in the scheme face competition from part-time drivers during off-peak periods. With an abundance of part-time drivers, the chances of an unprioritized driver receiving a service request during off-peak hours become extremely low, resulting in negligible utility. This similarity to a qualification scheme, where not participating also leads to zero off-peak utility, may lead one to believe that the platform sets the same peak period work target for both schemes. However, a comparison between Lemma 3 and Lemma 6 suggests that this intuition holds true only when the congestion cost is significant (i.e., $k > k_2$). In scenarios with a small congestion cost (i.e., $k \le k_2$), the platform sets a higher work target under the prioritization scheme compared with a qualification scheme.

To understand the reason, first note that to increase the matching volume under a qualification scheme, the platform seeks to maximize the utilization of full-time drivers' service. To achieve this, the platform shifts full-time drivers' work time to the peak period to minimize the likelihood of drivers experiencing idleness during the off-peak periods, as the supply of drivers may surpass the demand. With a small congestion cost, the platform using a qualification scheme is able to set a high peak work target that shifts enough full-time drivers' service to eliminate the off-peak supply idleness. However, under a prioritization scheme, all off-peak period service requests are fulfilled because of the abundant part-time drivers. Therefore, in contrast to that under a qualification system, the platform seeks to shift full-time drivers' work time to the peak period as much as possible. As a result, the platform is able to further increase the work target under a prioritization scheme because of the small congestion cost.

5.4. Comparison of the Benchmark and Two Schemes in an Open System

On comparing the benchmark scenario and the two incentive schemes in an open system, we have the following results:

Proposition 3. *In the open system, given $k > \frac{np\delta}{n-L+s}$ and that the price per unit of service time is fixed, we have the following results:*

- *Compared with the benchmark scenario, implementing a qualification scheme increases the platform's total matching volume if $k > k_4$ but decreases the total matching volume otherwise. Here,*

$$k_4 = \begin{cases} \dfrac{p\delta}{1-L/n}, & \text{if } s \le \sqrt{n^2-L^2}+L-n, \\ \tilde{k}, & \text{if } s > \sqrt{n^2-L^2}+L-n, \end{cases}$$

where $\tilde{k}$ satisfies $n - \int_{-s}^{n(1-\tilde{\alpha}_0^q)-L} F(\epsilon)d\epsilon - L - n\frac{p\delta}{\tilde{k}} = 0$, and $\tilde{\alpha}_0^q$ solves the implicit equation $\left(1-\frac{1}{n}\int_{-s}^{n(1-\tilde{\alpha}_0^q)-L}F(\epsilon)d\epsilon\right)p\delta - \frac{1}{2}\tilde{k}\tilde{\alpha}_0^{q2} = \frac{p^2\delta^2}{2\tilde{k}}$. In addition, it increases full-time drivers' welfare if $k \le k_2$ but has no impact on their welfare otherwise. Here, k_2 is defined in Lemma 3.

- *Compared with the benchmark scenario, implementing a prioritization scheme increases the platform's total matching volume without affecting full-time drivers' welfare.*
- *Compared with the qualification scheme, implementing a prioritization scheme (weakly) increases the platform's total matching volume. It lowers full-time drivers' welfare if $k \le k_2$ but otherwise does not affect their welfare.*

Recall that in a closed system, a qualification scheme always improves the total matching volume compared with the benchmark. Interestingly, Proposition 3 shows that in an open system, a qualification scheme may decrease the total matching volume. To see the reason, note that a qualification scheme, although it increases the peak period matching volume by shifting full-time drivers' services to the peak period, may also reduce the off-peak matching volume by rejecting part-time drivers from providing services. The total matching volume increases only when the congestion cost for full-time drivers is high. If the congestion cost is low, there is already an abundant supply of full-time drivers during the peak period when no qualification scheme is implemented. Consequently, the increase in peak period matching volume resulting from the supply shift in the scheme is outweighed by the loss of matching volume during the off-peak period because of the exclusion of part-time drivers.

Unlike a qualification scheme, a prioritization scheme improves the total matching volume (compared with the benchmark scenario) by shifting the supply of full-time drivers to the peak period while still allowing part-time drivers to work during the off-peak period. It also leads to a higher matching volume than a qualification scheme, in contrast to what we observe in a closed system where the prioritization scheme is inferior to the qualification scheme in increasing matching volume (see Proposition 1). This is because the prioritization scheme not only better utilizes the supply from part-time drivers but also enables the platform to set a higher peak period work target (as explained in Section 5.3), thereby effectively shifting more of the full-time drivers' supply to the peak period.

Proposition 3 further demonstrates the impacts of implementing incentive schemes on a full-time driver's welfare. In contrast to a closed system where the implementation of incentive schemes always hurts full-time drivers, a qualification scheme in an open system can benefit full-time drivers compared with not having any scheme. This happens when the congestion cost is small (i.e., $k \le k_2$). In such a case, full-time drivers are not significantly affected by longer working hours during the peak period, but they greatly benefit from reduced competition during off-peak periods because of the exclusion of part-time drivers under the qualification scheme. We also notice that threshold k_4 could be smaller than k_2. In this case, it implies a range of the congestion cost $k \in (k_4, k_2]$, under which a platform's implementing a qualification scheme benefits both the platform and full-time drivers (compared with not implementing any scheme).

We also observe that under certain situations, the welfare of full-time drivers is not affected by the implementation of either a qualification or a prioritization scheme. Note that in the absence of any scheme, a full-time driver's chances of receiving ride requests during off-peak periods are negligible. Consequently, the driver only derives positive utility during the peak period. When the platform uses either scheme to optimally shift full-time drivers' work time, it may set the work target such that the full-time drivers' gain from working in the off-peak period exactly offsets their loss because of the longer work time in the peak period, making the full-time drivers' welfare unaffected.

6. Model Extensions

In this section, we first compare the performance of the proposed incentive schemes to surge pricing in Section 6.1. We then relax our simplifying assumption by considering a finite number of part-time drivers in Section 6.2.[7]

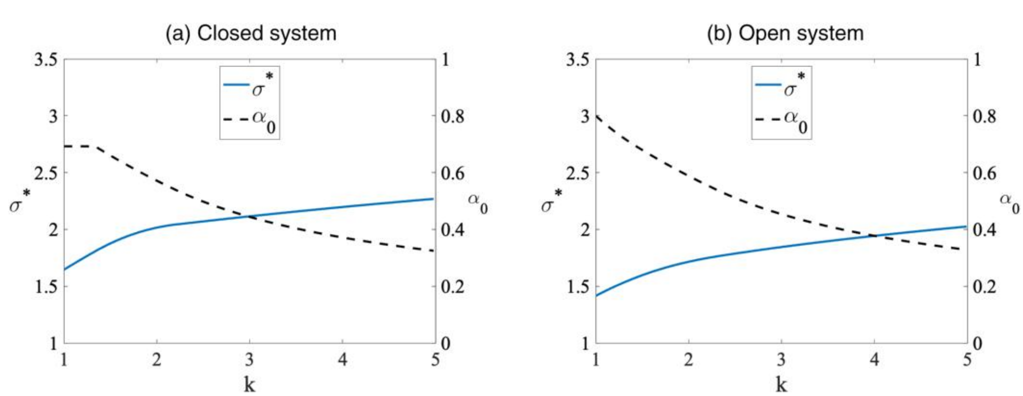
6.1. Incentive Schemes vs. Surge Pricing

A key motivation of our proposed incentive schemes is that prices in many markets can be subject to regulations and have limited usefulness in addressing the supply-demand imbalance. Nevertheless, it would be helpful to compare the performance of our proposed schemes with surge pricing, the practice of raising the transaction price during the peak period to address supply shortages. To model surge pricing, we let the peak period demand be $D_p - p_s$, where D_p denotes the base demand in the peak period (i.e., the demand for ride service when transaction price is zero), and p_s denotes the surging price in the peak period. The same as in the main model, p represents the regular price at the off-peak period. We define $\sigma = (p_s/p) > 1$ as the platform's peak period price multiplier, which reflects the level of price surging.

Given an exogenous price multiplier σ, we can analytically derive the total matching volume, the platform's profit, and drivers' utility when the platform uses surge pricing under both a closed system and an open system. The detailed analysis and results are provided in Online Appendix B. To compare with the incentive schemes, we calculate the minimum price multiplier needed for a surge pricing scheme to achieve the same profit as that by the optimal incentive scheme.[8] As the analytical comparison loses tractability, we resort to numerical analysis. The parameter values are carefully chosen based on several empirical studies, including Bai et al. (2019), Xiong et al. (2021), Bimpikis et al. (2019), and Wang and Zhang (2022). We provide the detailed methodologies in SA.D.1 of the supplementary analysis. We also carry out a series of robustness checks, varying one parameter at a time across 10 sets of parameter values. The detailed results can be found in SA.D.2 of the supplementary analysis.

Recall from Propositions 1 and 2 that the qualification scheme generates a higher matching volume in a closed system, whereas the prioritization scheme generates a higher matching volume in an open system. Thus, in the closed system, we analyze the price multiplier needed for surge pricing to match the same profit as the qualification scheme, whereas in the open system, we assess the price multiplier needed for surge pricing to match the prioritization scheme. Figure 1 shows that in both a closed system and an open system, given the parameter values and with a relatively small congestion cost (i.e., $k = 1$), the platform needs to raise its base price by approximately 50% for surge pricing to match the profit of the optimal incentive scheme. For this scenario (i.e., $k = 1$), the optimal scheme requires a work target around 0.7 (i.e., drivers need to allocate about 70% of their total daily work time to the peak period) in a closed system and around 0.8 (i.e., drivers need to allocate about 80% of their

Figure 1. (Color online) The Minimum Price Multiplier Needed for Surge Pricing to Achieve the Same Profit as an Incentive Scheme



Note. $D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$.

total daily work time to the peak period) in an open system.[9] As drivers' congestion cost grows larger, the multiplier required by surge pricing further increases, whereas the optimal incentive scheme's work target decreases. In the case of higher congestion costs such as $k = 5$, surge pricing in both systems requires almost doubling the price (i.e., $\sigma^* \approx 2$), whereas the optimal incentive scheme only requires the work target to be slightly above 0.3 in both systems.

As previously discussed, the proposed incentive schemes can serve as alternatives to surge pricing in markets where prices are regulated. Even in less regulated markets, implementing surge pricing with a higher price multiplier may face challenges. For example, a 50% price increase could draw criticism for unethical practices. The comparison suggests that as the congestion cost increases, implementing surge pricing becomes more challenging, whereas using the incentive scheme to achieve the same profits may become easier as the required work target decreases.

When deriving the optimal incentive schemes, the goal is to maximize the total matching volume because the price is exogenous. We now align the objective of surge pricing to also maximize the total matching volume and compare it with the optimal incentive scheme in each system. Interestingly, we find the possibility that surge pricing does not achieve the same matching volume as the incentive scheme. To understand the reason, note that whereas surge pricing increases the peak period service supply, it also decreases the peak period service demand, reducing the overall matching volume as a result. When the peak period demand is not significantly higher than the off-peak period demand, we could observe a lower total matching volume from surge pricing. Figure 2 illustrates this result in both a closed system and an open system.

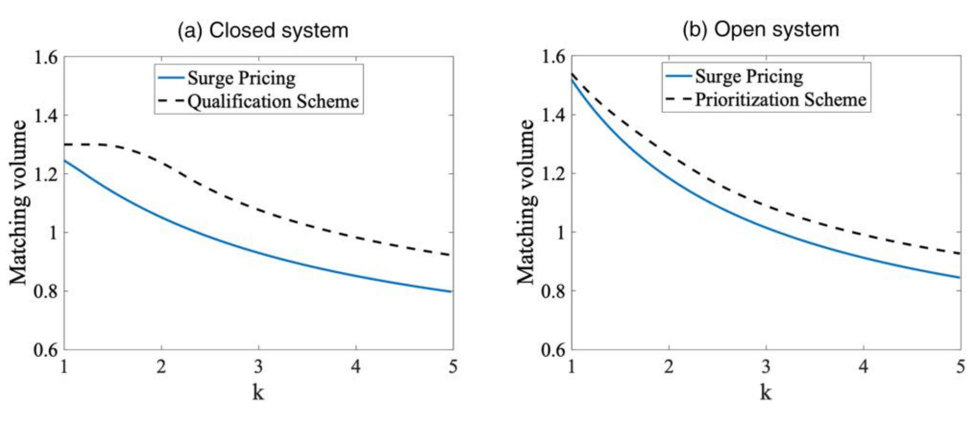
We further examine the mapping between the service target α_0 and the price multiplier σ such that the incentive scheme and surge pricing yield the same matching volume. In this analysis, the service target α_0 and the price multiplier σ are not necessarily the optimal ones but only enable the incentive scheme (i.e., the qualification scheme given a closed system and the prioritization scheme given an open system) and surge pricing to reach the same matching volume. Figure 3 presents the outcomes in both a closed system and an open system. For example, in a closed system and given the chosen parameter values, as the work target of the qualification scheme changes from 0.2 to 0.3, the price multiplier of surge pricing needs to change from around 1.4 (i.e., a 40% price increase) to two (i.e., a 100% price surge). If a high price multiplier becomes the obstacle in implementing surge pricing, the mapping result suggests that the proposed incentive schemes with reasonable work target may be feasible alternatives to achieve the same matching volume. Figure 3 also shows the region where surge pricing cannot achieve the same matching volume as in the incentive scheme, consistent with our observations in Figure 2.

In summary, the comparison suggests that our proposed schemes are viable alternatives to surge pricing, not only when prices are regulated but also when a high price multiplier is required but poses challenges in implementing surge pricing. Furthermore, as drivers' congestion cost grows, implementing surge pricing could become more difficult because of a higher price multiplier needed, whereas implementing an incentive scheme may become easier as the work target required decreases. The analysis also shows that the incentive schemes could have an advantage in improving the total matching volume.

6.2. Finite Part-Time Drivers

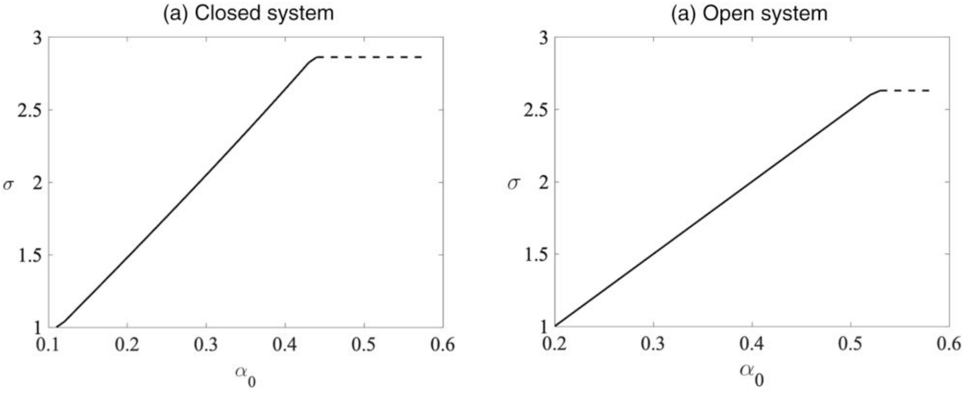
In the main model, we assumed that in an open system, an infinite number of part-time drivers can work during the off-peak period. The purpose of this assumption is to capture the supply surplus in the

Figure 2. (Color online) The Comparison of Maximum Matching Volume Between Surge Pricing and Incentive Schemes



Note. $D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, \epsilon \sim U[-0.1, 0.1]$.

Figure 3. The Minimum Price Multiplier Needed for Surge Pricing to Achieve/Approach the Matching Volume of an Incentive Scheme



Notes. (1) In the closed (open) system, surge pricing is compared with the qualification (prioritization) scheme. We use the "dashed" line to represent the case where the surge pricing fails to achieve the same matching volume as the incentive scheme. (2) In creating each figure, α_0 begins at the level that each full-time driver would choose voluntarily in the absence of an incentive scheme, corresponding to the case $\sigma = 1$ (i.e., no surge pricing). $D_0 = 1, D_p = 2, p = 0.5, \delta = 0.8, n = 1.3, k = 2, \epsilon \sim U[-0.1, 0.1]$.

off-peak period because of the involvement of part-time drivers in the simplest way and to study how this surplus enables an incentive scheme, particularly a prioritization scheme, to effectively incentivize full-time drivers to participate. We now relax this assumption by considering the case where the number of part-time drivers is finite and explore to what extent our result—that the prioritization scheme outperforms the qualification scheme in an open system (see Proposition 3)—continues to hold.

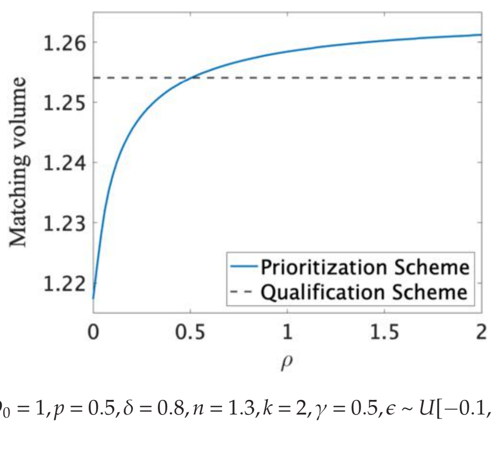
To explicitly incorporate a finite number of part-time drivers, we denote the ratio between part-time drivers and full-time drivers in an open system as ρ ($\rho > 0$). The total number of part-time drivers is then ρn. We denote the fraction of time per day that a part-time driver works as γ (note that we normalize the daily working hours of a full-time driver to one, so $\gamma < 1$). Therefore, each part-time driver provides a supply of γ per day, and the total daily supply from part-time drivers is $\rho n \gamma$.

Because part-time drivers are not allowed to provide service under a qualification scheme, the equilibrium characterization under this scheme remains unaffected when the number of part-time drivers is finite. However, the equilibrium characterization under a prioritization scheme depends on the number of available part-time drivers. Recall that in our main model, the advantage of a prioritization scheme over a qualification scheme stems from the inclusion of abundant part-time drivers. This inclusion allows the platform to benefit from further improving the work target under a prioritization scheme, even when the likelihood of full-time drivers experiencing idleness is eliminated. The next proposition shows that this result still holds as long as the number of part-time drivers exceeds a certain threshold.

Proposition 4. *In an open system with a finite number of part-time drivers, given that the price per unit of service time is fixed, a prioritization scheme leads to a higher matching volume than a qualification scheme as long as* ρ *is above a threshold (i.e.,* $\rho > \rho^*$*). The threshold* ρ^* *is obtained by solving* $M_O^{P\dagger}|_{\rho=\rho^*} = M_C^Q$*, where* M_C^Q *is given in Lemma 3 and* $M_O^{P\dagger}|_{\rho=\rho^*}$ *is provided in SA.B.1 of the supplementary analysis.*

We further visualize the comparison result in Figure 4. As the number of part-time drivers decreases, the benefit of allowing these part-time drivers to serve as backup supply to meet off-peak demand under the prioritization scheme diminishes. In the scenario where the number of part-time drivers becomes zero, the open system degenerates into a closed system. In such a case, the qualification scheme outperforms the prioritization scheme, consistent with our prediction in Proposition 1.

Figure 4. (Color online) Comparison of Total Matching Volume in an Open System with Finite Part-Time Drivers



Note. $D_0 = 1, p = 0.5, \delta = 0.8, n = 1.3, k = 2, \gamma = 0.5, \epsilon \sim U[-0.1, 0.1]$).

7. Conclusion

This study examines the challenge of intertemporal supply-demand imbalance faced by on-demand ride-hailing platforms. In contrast to existing literature that seeks demand-side solutions such as surge pricing, this study explores an operational approach and focuses on supply-side solutions. We propose two novel incentive schemes: the qualification scheme and the prioritization scheme. Under the qualification scheme, the platform sets a peak period work target and assigns off-peak requests only to drivers who meet the peak period work target. Under the prioritization scheme, the platform also sets a peak period work target but prioritizes the off-peak requests for drivers who meet the target. Our research finds that for platforms with a closed system that only allows full-time drivers, the qualification scheme is more effective in improving the total matching volume. However, it hurts full-time drivers more than the prioritization scheme. On the other hand, for platforms with an open system that also includes abundant part-time drivers to serve off-peak requests, the prioritization scheme outperforms the qualification scheme in improving total matching volume. Moreover, we find that the implementation of an incentive scheme in an open system may benefit both the platform and full-time drivers.

Managerially, our study offers an alternative to surge pricing for on-demand platforms to alleviate the intertemporal supply-demand imbalance. The findings are particularly useful for markets where prices are subject to regulation and inflexible to adjust. Besides China, India also regulates the price in the ride-hailing industry by capping surge pricing during peak demand at a maximum of 1.5 times the base fare (Dash 2020). In Karnataka, an Indian state, surge pricing is entirely prohibited (Siddiqui 2024). Similarly, the Philippine government limits surge multipliers to a maximum of two (Ilagan and Rainis 2024). Especially when a large price multiplier poses challenges in implementing surge pricing, we find that the proposed incentive schemes could have an implementation advantage. Furthermore, we show that the incentive schemes could potentially lead to a total matching volume that cannot be achieved by surge pricing. Our results provide useful insights for platforms with varying degrees of supply system openness on the choice of an incentive scheme. The welfare results offer guidance to platforms and policymakers who care about not only the number of matches made but also the welfare of independent contractors.

Our model can serve as a starting point for exploring other important issues in the broader gig economy. First, for the purpose of this study, we assume the composition of full-time and part-time workers is exogenous. Future research could explore the impact of implementing an incentive scheme on workers' strategic decisions to work full-time or part-time and thus the worker composition on the supply side. Second, one of our goals is to understand the effect of supply system openness on the performance of incentive schemes. Further work could incorporate relevant factors to explore the platform's choice of system openness. Third, although our paper examines two extremes of a platform's openness—which, in practice, usually exist on a continuous spectrum—the results of the closed system analysis apply to platforms that primarily rely on full-time drivers to provide service, whereas the results of the open system analysis apply to platforms that involve a large number of part-time drivers. For example, Proposition 4 indicates that the effectiveness of the two schemes hinges on the ratio of part-time drivers to full-time drivers, which could be used to measure the openness of a platform's supply system. Investigating a platform's optimal choice of supply system openness could be a promising direction for future research. Fourth, it would be beneficial to expand on the current work by examining the competition between platforms that can adopt different incentive schemes. Finally, empirical validation of our model predictions using field data would provide valuable insights.

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Endnotes

[1] For example, city governments in China can set government-guided prices for ride-hailing services and enforce them when they deem it necessary. See http://www.gov.cn/zhengce/2022-12/06/content_5730384.htm.

[2] See http://cloud.tencent.com/developer/news/572397.

[3] In Section SA.B.3 of the supplementary analysis, we demonstrate the robustness of our results when allowing a proportion of part-time drivers to provide service during peak periods.

[4] In our main model, we do not put a constraint on the length of peak and off-peak periods. That is, a full-time driver can allocate all his working time to either the peak or the off-peak period. Sometimes the peak period may be shorter than a full-time driver's total working time. We consider this situation by allowing the length of the peak period to be capped at $\overline{\alpha} \in (0,1)$. In this case, a full-time driver's choice needs to satisfy $\alpha \in [0,\overline{\alpha}]$. We show the robustness of our main results and present the analysis in SA.B.5 of the supplementary analysis.

[5] The detailed analysis can be found in the SA.B.3 of the supplementary analysis.

[6] Although not explicitly modeled, the demand during peak periods can also be volatile. Our results, however, remain unaffected as long as the peak demand always exceeds the service supply ($D_p(p) > n$). In SA.B.4 in the supplementary analysis, we analyze the scenario in which the peak period demand exceeds n with certain probability and demonstrate the robustness of our results.

[7] In Online Appendix C, we further incorporate drivers' heterogeneity in their congestion costs and discuss the robustness and boundary conditions of our results. In SA.D.5 of the supplementary analysis, we conducted additional robustness checks by considering drivers' heterogeneous congestion costs.

[8] In SA.D.4 of the supplementary analysis, we use an alternative comparison approach by comparing surge pricing and the incentive schemes when they are all maximizing the platform's profits. With its full flexibility in adjusting prices, it is not surprising that surge pricing yields the highest platform profit. However, we find that the incentive scheme generates a higher matching volume and increases consumer surplus.

[9] In reality, the length of the peak period may be limited to less than one so that a worker cannot allocate its full time to the peak period. In SA.D.3 of the supplementary analysis, we analyze cases where the service target has upper bounds of 0.5, 0.4, and 0.3, respectively. We show that the result of comparison with surge pricing is robust as long as drivers' congestion costs are high.

References

Afèche P, Liu Z, Maglaras C (2023) Ride-hailing networks with strategic drivers: The impact of platform control capabilities on performance. *Manufacturing Service Oper. Management* 25(5):1890–1908.

Ashkrof P, de Almeida Correia GH, Cats O, van Arem B (2020) Understanding ride-sourcing drivers' behaviour and preferences: Insights from focus groups analysis. *Res. Transportation Bus. Management* 37:100516.

Bai J, So KC, Tang CS, Chen X, Wang H (2019) Coordinating supply and demand on an on-demand service platform with impatient customers. *Manufacturing Service Oper. Management* 21(3):556–570.

Benjaafar S, Hu M (2020) Operations management in the age of the sharing economy: What is old and what is new? *Manufacturing Service Oper. Management* 22(1):93–101.

Besbes O, Castro F, Lobel I (2021) Surge pricing and its spatial supply response. *Management Sci.* 67(3):1350–1367.

Besbes O, Goyal V, Iyengar G, Singal R (2024) Workforce scheduling with heterogeneous time preferences: Effective wages and workers' supply. *Manufacturing Service Oper. Management* 26(5):1768–1786.

Bimpikis K, Candogan O, Saban D (2019) Spatial pricing in ride-sharing networks. *Oper. Res.* 67(3):744–769.

Cachon GP, Daniels KM, Lobel R (2017) The role of surge pricing on a service platform with self-scheduling capacity. *Manufacturing Service Oper. Management* 19(3):368–384.

Castro F, Gao J, Martin S (2022) Supply prioritization in hybrid marketplaces. Preprint, submitted June 7, http://dx.doi.org/10.2139/ssrn.4119096.

Dash S (2020) Uber, Ola hit a speed bump as Indian government caps surge pricing and issues new guidelines. *Econom. Times* (November 27), https://economictimes.indiatimes.com/tech/startups/uber-ola-hit-a-speed-bump-as-indian-government-caps-surge-pricing-and-issues-new-guidelines/articleshow/79443945.cms?from=mdr.

Evans GW, Wener RE, Phillips D (2002) The morning rush hour: Predictability and commuter stress. *Environment Behav.* 34(4):521–530.

Feng G, Kong G, Wang Z (2021) We are on the way: Analysis of on-demand ride-hailing systems. *Manufacturing Service Oper. Management* 23(5):1237–1256.

Feng H, Feng N, Zhang L, Jiang Z, Li M (2025) A study of ride-hailing platforms' business models in the presence of surge demand. *Inform. Systems Res.*, ePub ahead of print March 13, https://doi.org/10.1287/isre.2022.0695.

Goh B, Shirouzu N (2018) Pressures on Didi intensify after latest passenger killing. *Reuters* (August 30), https://www.reuters.com/article/technology/idUSKCN1LF00V/.

Gulian E, Debney LM, Glendon AI, Davies DR, Matthews G (1989) Coping with driver stress. McGuigan FJ, Sime WE, Wallace JM, eds. *Stress and Tension Control 3* (Springer, Boston), 173–186.

Hennessy DA, Wiesenthal DL (1997) The relationship between traffic congestion, driver stress and direct versus indirect coping behaviours. *Ergonomics* 40(3):348–361.

Hennessy DA, Wiesenthal DL (1999) Traffic congestion, driver stress, and driver aggression. *Aggressive Behav.* 25(6):409–423.

Hu M, Zhou Y (2022) Dynamic type matching. *Manufacturing Service Oper. Management* 24(1):125–142.

Hu B, Hu M, Zhu H (2022) Surge pricing and two-sided temporal responses in ride hailing. *Manufacturing Service Oper. Management* 24(1):91–109.

Hu M, Wang J, Zhang ZJ (2022) Implications of worker classification in on-demand economy. Preprint, submitted April 18, http://dx.doi.org/10.2139/ssrn.4076484.

Huang Y, Sun DJ, Zhang L-H (2018) Effects of congestion on drivers' speed choice: Assessing the mediating role of state aggressiveness based on taxi floating car data. *Accident Anal. Prevention* 117:318–327.

Ilagan K, Rainis FA (2024) How PCIJ investigated grab's surge pricing model. *PCIJ* (July 12), https://pcij.org/2024/07/12/methodology-pcij-investigation-grab-surge-pricing-model/.

Kapoor A, Tucker CE (2017) How do platform participants respond to an unfair rating? An analysis of a ride-sharing platform using a quasi-experiment. Preprint, submitted May 19, http://dx.doi.org/10.2139/ssrn.2970772.

Liao R (2019) The end of China's ridesharing gig. *TechCrunch* (January 1), https://techcrunch.com/2019/01/01/end-of-chinas-ride-sharing-gig/.

Liu C, Lu Y (2021) Modelling and optimization for the employment of full-time and part-time drivers in ride-hailing platforms. *Internat. J. Econom. Finance Management Sci.* 9(6):221–224.

Morris EA, Hirsch JA (2016) Does rush hour see a rush of emotions? Driver mood in conditions likely to exhibit congestion. *Travel Behav. Soc.* 5:5–13.

Narasimhan C, Papatla P, Jiang B, Kopalle PK, Messinger PR, Moorthy S, Proserpio D, Subramanian U, Wu C, Zhu T (2018) Sharing economy: Review of current research and future directions. *Customer Needs Solutions* 5(1):93–106.

Nonninger L (2019). Chinese ride-hailing app Didi Chuxing is rolling out financial services. *Bus. Insider* (January 3), https://www.businessinsider.com/didi-launches-user-insurance-loans-2019-1.

OFWeek (2023) Online ride-hailing services are in an awkward period, with many passengers complaining and drivers unable to receive orders. *OFWeek* (June 25), https://iot.ofweek.com/2023-06/ART-132215-8500-30600985.html.

Shen J (2019) Didi raises fares in Beijing in a bid to ease driver shortages. *Technode* (July 9), https://technode.com/2019/07/09/didi-price-increase-beijing/.

Siddiqui M-UR (2024) In a first, Karnataka brings in uniform fares for all cabs. *Deccan Herald* (February 4), https://www.deccanherald.com/india/karnataka/2879202.

Sohu (2021) Tianjin Licheng special car driver assessment standards for April released, failure to meet the standards will be demoted to time-sharing drivers. *Sohu* (April 2), https://www.sohu.com/a/458602481_442726.

Stokols D, Novaco RW, Stokols J, Campbell J (1978) Traffic congestion, type a behavior, and stress. *J. Appl. Psych.* 63(4):467–480.

Tang S (2018) Geely-backed EV ride-hailing firm closes USD150 million A-round. *YiCai Global* (January 17), https://www.yicaiglobal.com/news/geely-backed-ev-ride-hailing-firm-closes-usd150-million-a-round.

Tang Y, Guo P, Tang CS, Wang Y (2021) Gender-related operational issues arising from on-demand ride-hailing platforms: Safety concerns and system configuration. *Production Oper. Management* 30(10):3481–3496.

Tang Y, Yang H, Wang B, Huang J, Bai Y (2019) A Pareto-improving and revenue-neutral scheme to manage mass transit congestion with heterogeneous commuters. *Transportation Res. Part C Emerging Tech.* 113:245–259.

Taylor TA (2018) On-demand service platforms. *Manufacturing Service Oper. Management* 20(4):704–720.

Wang F (2019) Experts question the dynamic pricing mechanism for ride-hailing services during peak hours: A monopoly that is extremely unbalanced. *21st Century Bus. Herald* (September 25), https://m.21jingji.com/article/20190925/herald/4e1b5945e41240ea095b13f45e970356.html.

Wang H, Yang H (2019) Ridesourcing systems: A framework and review. *Transportation Res. Part B Methodological* 129:122–155.

Wang X, Zhang R (2022) Carpool services for ride-sharing platforms: Price and welfare implications. *Naval Res. Logist.* 69(4): 550–565.

Wei J (2019) Didi changes pricing rules in Beijing. *China Daily* (July 10), https://www.chinadaily.com.cn/a/201907/10/WS5d257efaa3105895c2e7cb93.html.

Xiong Z, Li J, Wu H (2021) Understanding operation patterns of urban online ride-hailing services: A case study of Xiamen. *Transport Policy* 101:100–118.

Xu L, Wei Y, Zhu Y (2024) Customer voice on two-sided platforms: The effect of surge pricing on customer complaints. *Management Sci.*, ePub ahead of print November 18, https://doi.org/10.1287/mnsc.2024.4992.

Yang H, Tang Y (2018) Managing rail transit peak-hour congestion with a fare-reward scheme. *Transportation Res. Part B Methodological* 110:122–136.

Yang H, Shao C, Wang H, Ye J (2020) Integrated reward scheme and surge pricing in a ridesourcing market. *Transportation Res. Part B Methodological* 134:126–142.

Yu JJ, Tang CS, Max Shen Z-J, Chen XM (2020) A balancing act of regulating on-demand ride services. *Management Sci.* 66(7):2975–2992.